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10. Write a Python Program to Implement Expectation & Maximization Algorithm

1. Student Performance Clustering

Dataset: student.csv

Study Hours	Marks
2	40
3	45
2.5	42
7	80
8	85
7.5	82

PROGRAM

```
import numpy as np
from sklearn.mixture import GaussianMixture
X = np.array([
    [2,40],[3,45],[2.5,42],
    [7,80],[8,85],[7.5,82]
])
model = GaussianMixture(n_components=2, random_state=0)
model.fit(X)
print("Cluster Labels:")
print(model.predict(X))
```

```
print("Means:")  
print(model.means_)
```

OUTPUT

Cluster Labels:

```
[1 1 1 0 0 0]
```

Means:

```
[[ 7.5  82.33]
```

```
 [ 2.5  42.33]]
```

2. Customer Spending Clustering

Dataset: customer.csv

Age	Spending
20	200
22	250
25	300
40	800
42	850
45	900

PROGRAM

```
import numpy as np  
from sklearn.mixture import GaussianMixture  
X = np.array([  
    [20,200],[22,250],[25,300],  
    [40,800],[42,850],[45,900]  
])  
em = GaussianMixture(n_components=2, random_state=1).fit(X)
```

```
print("Labels:", em.predict(X))  
print("Cluster Means:")  
print(em.means_)
```

OUTPUT

Labels:

```
[1 1 1 0 0 0]
```

Cluster Means:

```
[[ 42.33 850. ]
```

```
 [ 22.33 250. ]]
```

3. Height and Weight Clustering

Dataset: height.csv

Height	Weight
150	50
155	52
160	55
175	72
180	76
185	80

PROGRAM

```
import numpy as np  
from sklearn.mixture import GaussianMixture  
X = np.array([  
    [150,50],[155,52],[160,55],  
    [175,72],[180,76],[185,80]  
])
```

```
gmm = GaussianMixture(n_components=2, random_state=2)
gmm.fit(X)
print("Clusters:", gmm.predict(X))
print("Means:")
print(gmm.means_)
```

OUTPUT

Clusters:

```
[1 1 1 0 0 0]
```

Means:

```
[[180.  76. ]
```

```
 [155.  52.33]]
```

4. House Area and Price Clustering

Dataset: house.csv

Area	Price
600	30
700	35
800	40
1500	75
1600	82
1700	88

PROGRAM

```
import numpy as np
from sklearn.mixture import GaussianMixture
X = np.array([
    [600,30],[700,35],[800,40],
```

```
[1500,75],[1600,82],[1700,88]
])
model = GaussianMixture(n_components=2, random_state=3)
model.fit(X)
print("Cluster:", model.predict(X))
print("Means:")
print(model.means_)
```

OUTPUT

```
Cluster:
[1 1 1 0 0 0]
Means:
[[1600.  81.67]
 [ 700.  35.  ]]
```

5. Temperature and Humidity Clustering

Dataset: weather.csv

Temperature	Humidity
20	80
22	78
24	75
32	45
35	40
38	35

PROGRAM

```
import numpy as np
from sklearn.mixture import GaussianMixture
```

```
X = np.array([
    [20,80],[22,78],[24,75],
    [32,45],[35,40],[38,35]
])
em = GaussianMixture(n_components=2, random_state=4)
em.fit(X)
print("Cluster Labels:")
print(em.predict(X))
print("Cluster Means:")
print(em.means_)
```

OUTPUT

Cluster Labels:

```
[1 1 1 0 0 0]
```

Cluster Means:

```
[[35.  40. ]
```

```
 [22.  77.67]]
```